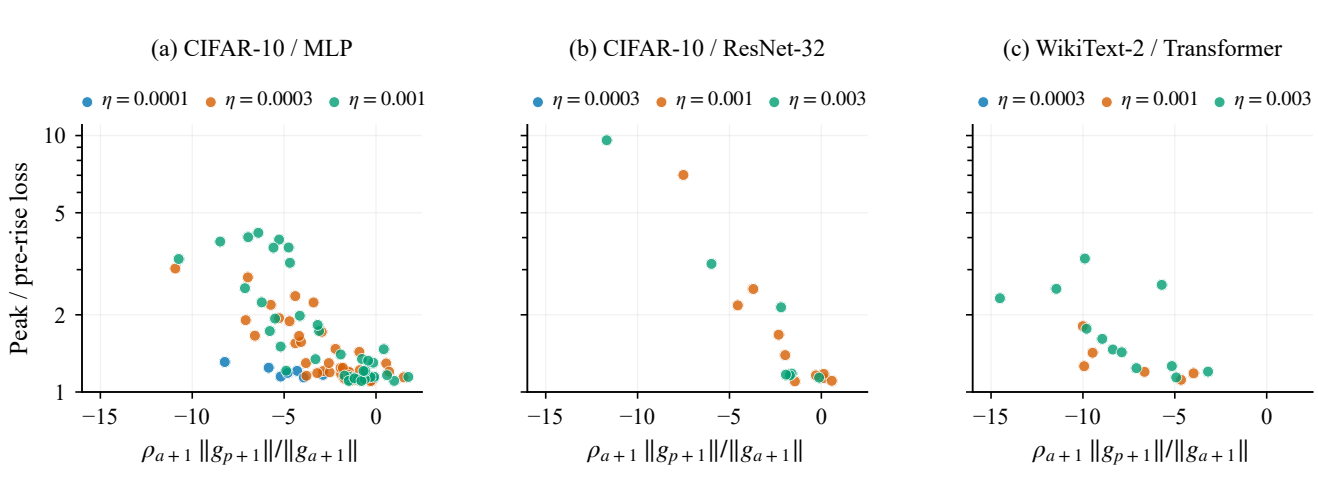

(a) CIFAR-10 / MLP
(b) CIFAR-10 / ResNet-32
(c) WikiText-2 / Transformer
η = 0.0001
η = 0.0003
η = 0.001
η = 0.003
Peak / pre-rise loss
10
5
2
1
−15
−10
−5
0
ρa+1 ‖gp+1‖/‖ga+1‖